

BEYOND NODAL DAMPING

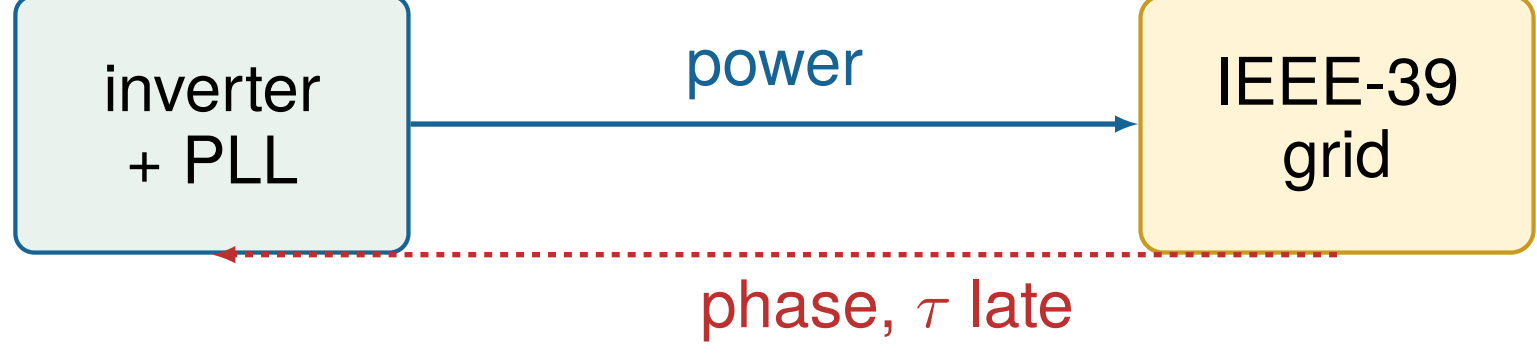
PRESERVING ONE MODE DOES NOT PROTECT THE GRID

Exact PLL delay compensation and its network-wide price · IEEE-39

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1 A PLL THAT LISTENS LATE

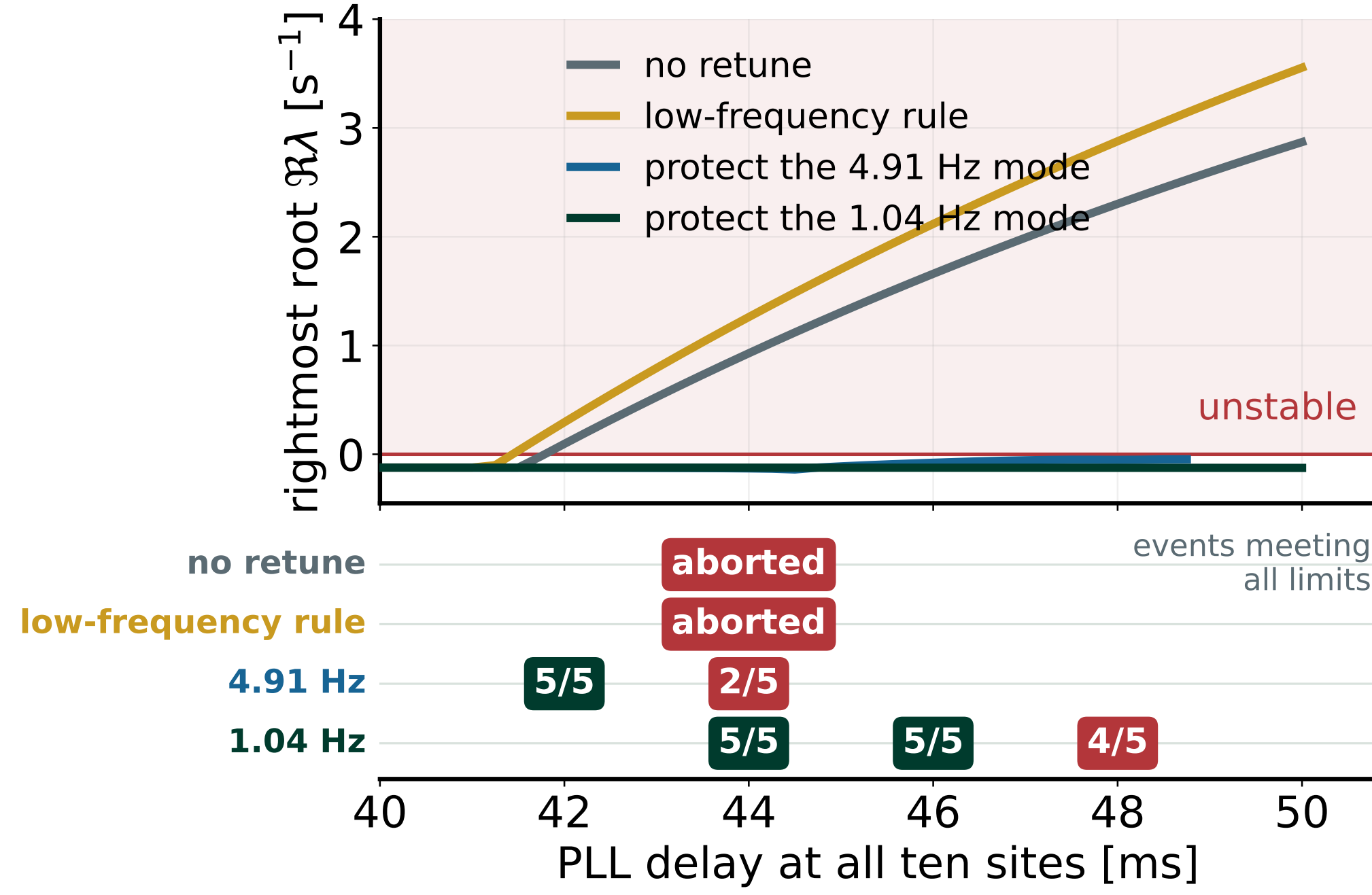
Inverters replace synchronous generators. Each one tracks the grid phase with a phase-locked loop (PLL), and that measurement arrives τ late.



If a retune keeps one oscillation mode exactly where it was, is the grid safe?

IEEE-39 at 87.5% inverter share, ten PLLs. Their delay grows from 40 to 44 ms.

2 THE USUAL COMPENSATION MAKES IT WORSE



12 vs 0

unstable roots at 44 ms: the low-frequency rule against protecting the 1.04 Hz mode.

IN WORDS Which mode you protect decides everything. The low-frequency rule is worse than no retune. Protecting the 1.04 Hz mode keeps the spectrum and passes 5 of 5 events.

3 TWO GAINS, ONE MODE

PLL loop factor $\kappa(s) = (k_p s + k_I) e^{-s\tau}$. The delay grows by h ; we protect the mode $\lambda = \alpha + j\omega$.

Exact transport of the gains

$$k_p(h)\lambda + k_I(h) = (k_p\lambda + k_I) e^{\lambda h}$$

What is left at every other frequency

$$\Delta\kappa(s) = -(s-\lambda)(s-\bar{\lambda}) e^{-s\tau} \int_0^h k_p(t) e^{-st} dt$$

IN WORDS Zero error at the protected mode, quadratic error everywhere else. The gains need no network model.

1 DELAY GROWS

$$\tau \rightarrow \tau + h$$

2 TRANSPORT

$$k\lambda \rightarrow k\lambda e^{\lambda h}$$

3 SPILLOVER

$$(\mu - \lambda)(\mu - \bar{\lambda}) \partial_{k_I} \mu$$

4 CHOOSE AND CHECK

spectrum and events

4 WHY: THE NETWORK SETS THE PRICE

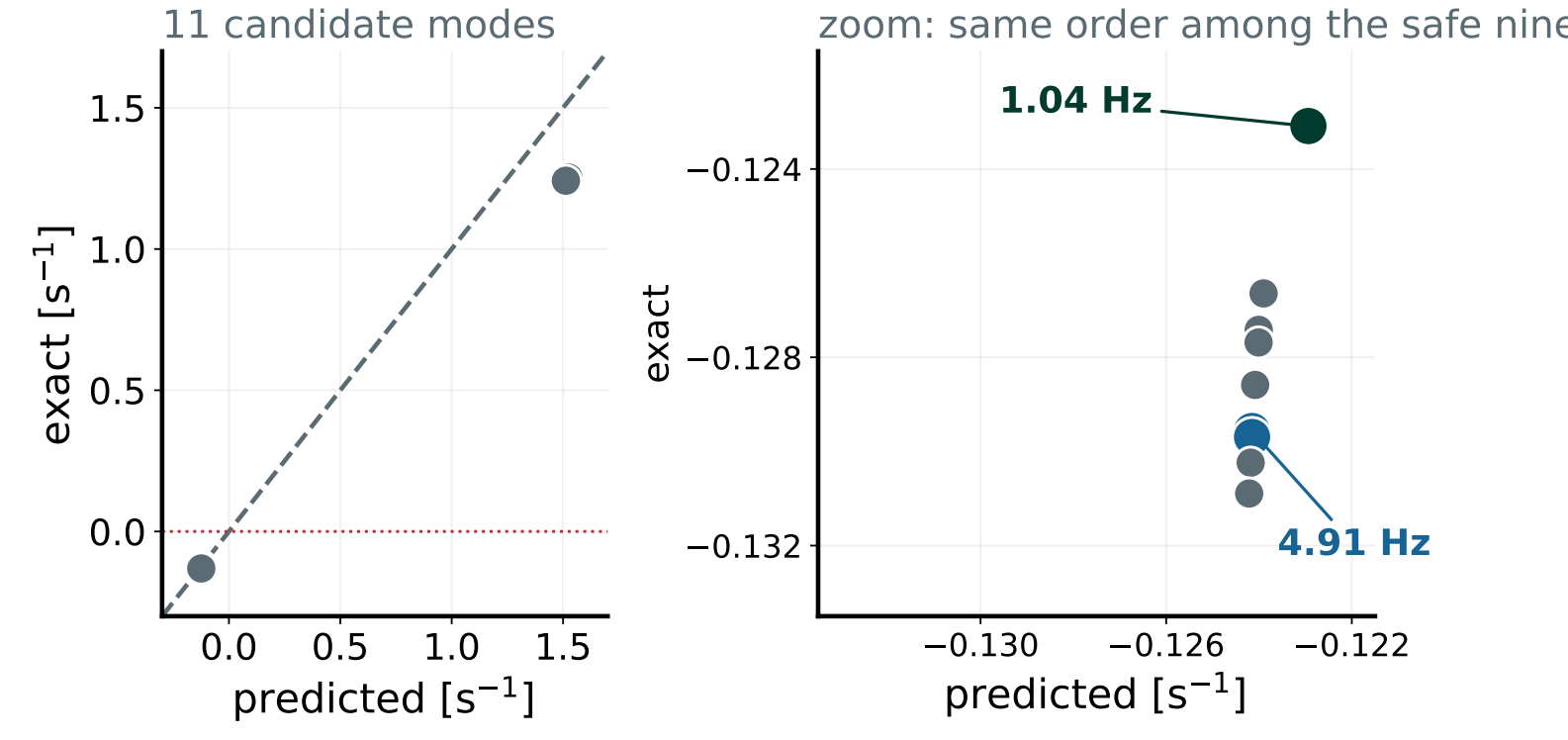
Spillover onto any other mode μ

$$\frac{d\mu}{dh} = -k_p(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial\mu}{\partial k_I}$$

The PLL fixes the quadratic factor. The network fixes the residue $\partial\mu/\partial k_I$, through its reduced operator $Z(s)$.

0.99 rank correlation between predicted and exact price over 11 candidate modes.

One analysis ranks the choices

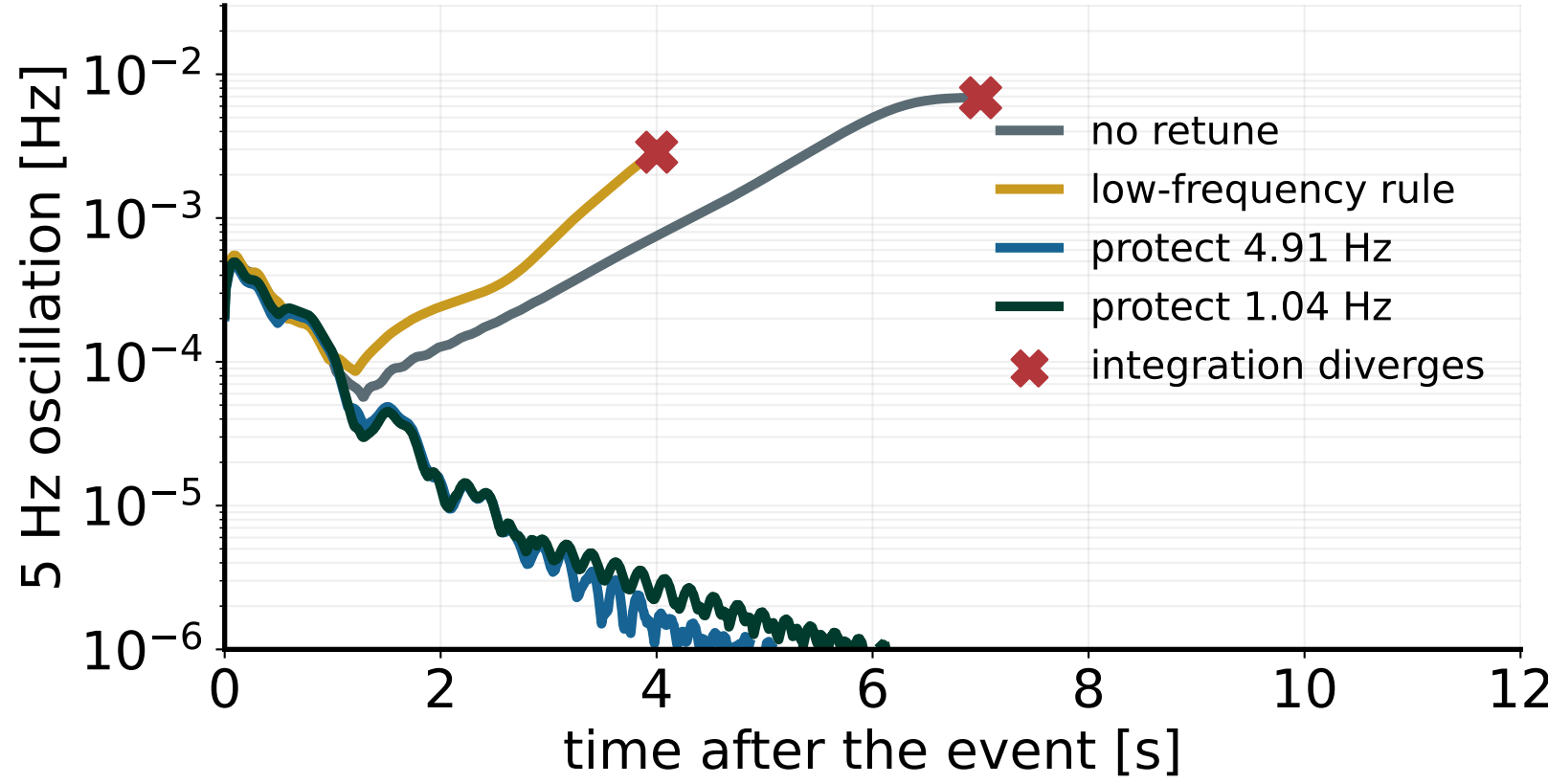


Worst root after +4 ms when protecting each mode.

The selection rule

- 1 One eigen-analysis of the base design.
- 2 Predict the worst root for each candidate mode.
- 3 Keep those inside the margin; take the one that keeps the most k_I .
- 4 Verify: root count and nonlinear events.

5 IN TIME: WHAT THE GRID DOES



The 5 Hz oscillation grows without a retune and with the low-frequency rule until the integration diverges. With either protected mode it dies out.

Bus 16 +100 MW step, 44 ms at all ten PLLs. 4.2–5.6 Hz band of the bus frequencies, largest of 39 buses; below 10^{-6} Hz not shown. Delayed nonlinear DAE.

6 SPECTRUM AND EVENTS: PASS BOTH

Roots clean, events fail

Protecting the 4.91 Hz mode at 44 ms leaves 0 roots beyond the margin, yet only **2 of 5** events meet the limits (0.505 Hz; SG reserve ≈ 0). The retune spends integral gain: k_I falls from 246.7 to 137.8.

Events clean, roots unstable

With no retune at 42 ms there are 4 unstable roots ($+0.096 \text{ s}^{-1}$ at 4.8 Hz), yet the event meets every limit for 30 s. The 0.5 s windows do not see a slow 5 Hz growth.

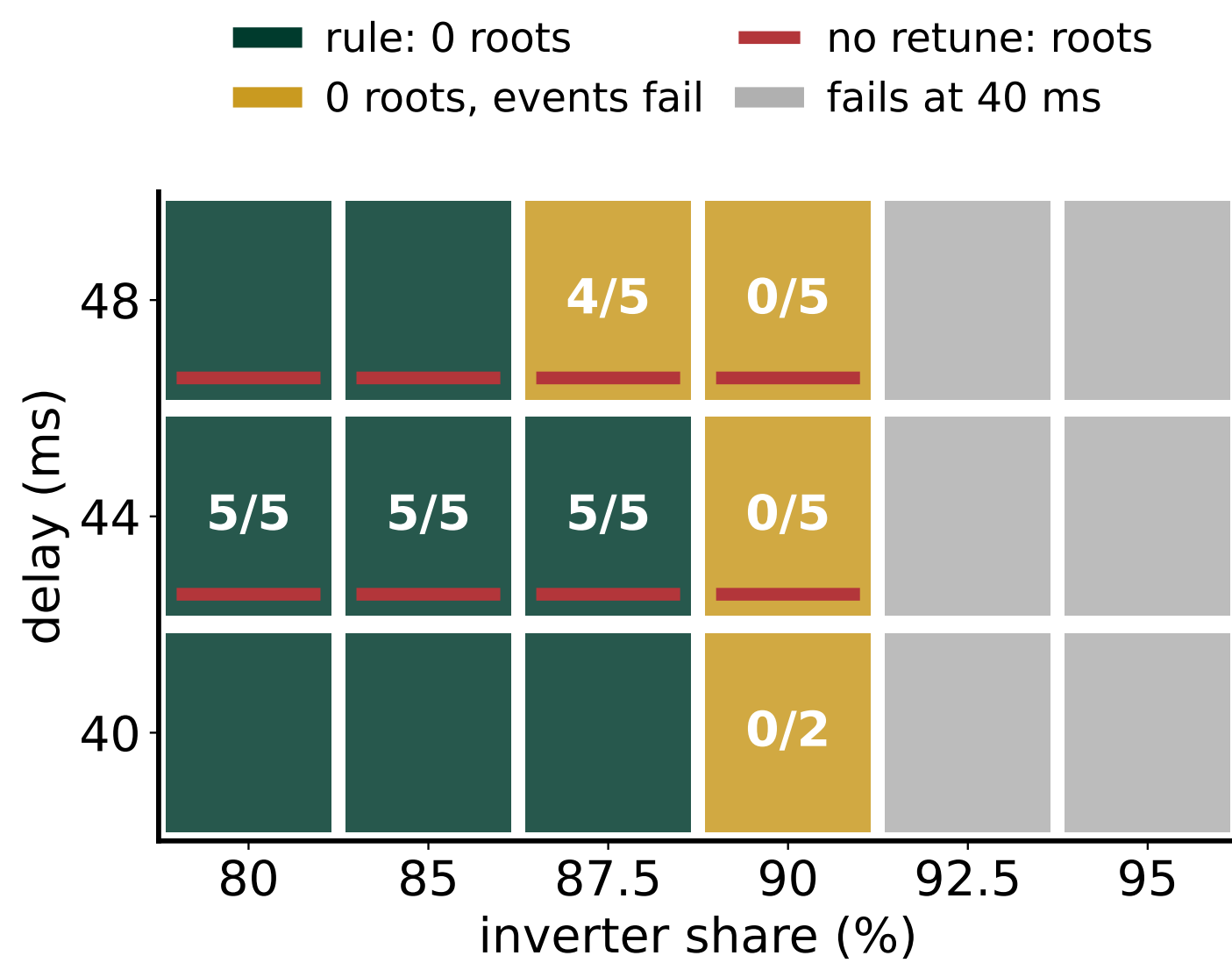
IN WORDS Neither test is enough alone: the spectrum misses the spent reserve, the events miss slow growth.

7 IT EXTENDS DELAY TOLERANCE, NOT THE MAXIMUM SHARE

Each cell retunes the PLLs by the rule and counts roots beyond the margin. A red bar marks the same cell without retuning. Numbers are nonlinear events passed.

IN WORDS Where the 40 ms design is feasible, the rule keeps the margin at 44 and 48 ms. 92.5% already fails at 40 ms.

At 90% even the 40 ms design fails the events (0 of 2): the limit there is the share, not the retune.



8 DOES IT TRANSFER? AND WHAT WE DO NOT CLAIM

5 of 5

events pass on a second base design (85% inverter share) with the rule applied unchanged: 0 roots beyond the margin.

Nine-bus bank (assumed dynamic data): transport on its three PLLs gives -0.131 s^{-1} from $+0.886$ (spectral re-run; its 12 of 12 events are reported).

First-order version of the formula gives the same root counts: the choice of mode matters, not the truncation.

Phase budget $h^* = \varphi_{PI}(\omega)/\omega$: 8.99 ms for the 4.91 Hz mode, a limit of the formula, not of operation.

IN WORDS Not claimed: an optimal mode, a maximum delay, a maximum inverter share, superiority over other compensators. Root counts are floating point, checked against eigenvalues at zero delay, not interval-certified. One network.

9 THREE RULES FOR THE ENGINEER

- 1 **Do not** compensate PLL delay at low frequency.
- 2 Choose the protected mode with the spillover law: one modal analysis is enough.
- 3 Check the **spectrum and the events**.

TAKEAWAY

PRESERVING ONE MODE DOES NOT PROTECT THE GRID.

The gains that hide a delay from one mode are exact. The network decides what they do to all the others.

REFERENCES

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- [2] Huang et al., arXiv:1903.05489.
- [3] Michiels & Gumussoy, arXiv:2003.05496.
- [4] Dörfler et al., IEEE Trans. Power Syst., 2014.
- [5] Bindel & Hood, SIAM J. Matrix Anal. Appl., 2013.
- [6] Johansson, IEEE Trans. Comput., 2017.

Topics: [1] converter small-signal stability; [2] PLL stability and grid structure; [3] fixed-order design with delays; [4] sparse wide-area control; [5] nonlinear eigenvalues; [6] interval arithmetic.

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